# Discussion: Identifiability and Optimization in Diagonal Kalman Delta Networks

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## 1 Setup

The diagonal Kalman Delta Network (KDN) predicts a diagonal covariance and uses it to form an uncertainty-dependent delta-rule write. For key $\boldsymbol{k}_t$, transition $\boldsymbol{\alpha}_t$, process noise $\boldsymbol{\omega}_t$, and observation noise $r_t$,

$$\widehat{\boldsymbol{p}}_t = \boldsymbol{\alpha}_t^2 \odot \boldsymbol{p}_{t-1} + \boldsymbol{\omega}_t, \tag{1}$$

$$\boldsymbol{\kappa}_t = \frac{\widehat{\boldsymbol{p}}_t \odot \boldsymbol{k}_t}{r_t + \sum_i \widehat{p}_{t,i} k_{t,i}^2}. \tag{2}$$

The memory update remains a delta update,

$$\boldsymbol{S}_t = \boldsymbol{D}_t \boldsymbol{S}_{t-1} + \boldsymbol{\kappa}_t \left( \boldsymbol{v}_t - (\boldsymbol{D}_t \boldsymbol{S}_{t-1})^\top \boldsymbol{k}_t \right)^\top, \qquad \boldsymbol{D}_t = \operatorname{diag}(\boldsymbol{\alpha}_t). \tag{3}$$

The implementation additionally tracks diagonal information $\boldsymbol{c}_t = 1/\boldsymbol{p}_t$ with the calibrated projected update

$$\boldsymbol{c}_t = \frac{1}{\widehat{\boldsymbol{p}}_t} + \frac{d_k}{r_t} \boldsymbol{k}_t^2. \tag{4}$$

This is a scan-compatible diagonal-information approximation, not an exact diagonal Kalman posterior.

## 2 The Scale Symmetry

For any constant $\lambda > 0$, consider

$$\boldsymbol{p}'_t = \lambda \boldsymbol{p}_t, \qquad \boldsymbol{\omega}'_t = \lambda \boldsymbol{\omega}_t, \qquad r'_t = \lambda r_t, \qquad \mu' = \frac{\mu}{\lambda}, \tag{5}$$

where $\mu = \boldsymbol{c}_0$ is the initial information and hence $\boldsymbol{p}_0 = 1/\mu$. The predicted covariance scales as

$$\widehat{\boldsymbol{p}}'_t = \boldsymbol{\alpha}_t^2 \odot (\lambda \boldsymbol{p}_{t-1}) + \lambda \boldsymbol{\omega}_t = \lambda \widehat{\boldsymbol{p}}_t. \tag{6}$$

Consequently, the gain is invariant:

$$\boldsymbol{\kappa}'_t = \frac{\lambda \widehat{\boldsymbol{p}}_t \odot \boldsymbol{k}_t}{\lambda r_t + \sum_i \lambda \widehat{p}_{t,i} k_{t,i}^2} = \boldsymbol{\kappa}_t. \tag{7}$$

The information update also transforms consistently as $\boldsymbol{c}'_t = \boldsymbol{c}_t/\lambda$. Therefore $\boldsymbol{\omega}$, $r$, and $\mu$ contain a common covariance-scale degree of freedom that cannot be identified from the memory output. Positive floors weaken the symmetry near their boundaries but do not remove the general conditioning problem.

The transition $\boldsymbol{\alpha}_t$ is different. It affects both covariance prediction and the memory transition $\boldsymbol{D}_t\boldsymbol{S}_{t-1}$, so it is not part of the exact scale symmetry. Nevertheless, $\boldsymbol{\alpha}_t$ and $\boldsymbol{\omega}_t$ are partially confounded in the covariance path because multiple pairs can produce similar $\widehat{\boldsymbol{p}}_t$.

# 3 What the Noisy-MQAR Results Establish

The original diagonal model obtains $76.2 \pm 22.3$ percent over the $5 \times 5$ initialization and data grid. Its initialization component of variation (19.2 points) is larger than its data component (11.4 points). This is not uniform under-capacity: many cells reach 88–96 percent, while a few fail severely.

The most informative intervention is isotropic gain initialization. Zeroing the output map of the process-noise projection makes $\boldsymbol{\omega}_t$ uniform across channels at initialization, after which anisotropy is learned. At the same learning rate and with the same optimizer, this changes performance from

$$76.2 \pm 22.3 \quad \longrightarrow \quad 83.7 \pm 10.4, \tag{8}$$

and halves the initialization variation from 19.2 to 9.0 points. It also rescues both severe collapse cells. Thus the current evidence identifies random initial gain anisotropy as the primary failure, not the nominal learning rate or scale symmetry.

The scale symmetry remains relevant as a secondary optimization defect. It creates a flat direction, gives equivalent physical models different raw parameter values, and lets coupled quantities evolve under different optimizer geometries. In the current Muon setup, matrix weights use Muon with adjusted learning rates, while biases and the one-dimensional prior parameter use AdamW. A single printed learning rate therefore does not imply uniform motion of the physical uncertainty variables.

The prior $\mu$ is unlikely to explain late failures. It only controls the initial information, and its influence should decay rapidly over a long sequence when process noise has a positive floor. Learning it adds a weakly identified parameter with little expected benefit.

# 4 Ways to Remove or Reduce the Symmetry

## 4.1 Fix the observation-noise scale

The strongest and simplest gauge choice is

$$r_t \equiv 1, \qquad \mu \equiv 1. \tag{9}$$

The model then learns process uncertainty in units of observation noise:

$$\boldsymbol{\omega}_t = \omega_{\min} + \operatorname{softplus}(f_\omega(\boldsymbol{x}_t)). \tag{10}$$

Because $r$ can no longer be rescaled, the common covariance-scale symmetry is removed. This parameterization is the preferred baseline unless token-dependent observation reliability is shown empirically to matter.

### 4.2 Learn bounded relative observation reliability

If heteroscedastic observation noise is useful, it should change confidence relative to a fixed reference rather than learn an unrestricted global scale. One option is

$$r_t = \exp(s \tanh f_r(\boldsymbol{x}_t)), \tag{11}$$

with fixed $s$. For $s = \log 4$, $r_t \in [1/4, 4]$. The projection should have no learnable global bias, and the prior can remain fixed at $\mu = 1$. A soft anchoring penalty can further control drift,

$$\mathcal{L}_{\text{scale}} = \gamma \left(\mathbb{E}[\log r_t]\right)^2. \tag{12}$$

This reduces rather than mathematically eliminates every possible input-dependent rescaling, but it prevents the practically harmful unbounded common-scale direction.

### 4.3 Parameterize a stationary uncertainty level

The $\boldsymbol{\alpha}$–$\boldsymbol{\omega}$ interaction can be made more interpretable with

$$\omega_{t,i} = \omega_{\min} + (1 - \alpha_{t,i}^2)\rho_{t,i}, \qquad \rho_{t,i} = \text{softplus}(f_\rho(\boldsymbol{x}_t)). \tag{13}$$

Here $\alpha$ controls retention and $\rho$ is the uncertainty level toward which prediction moves. This is exactly the completed `DiagKLA1` experiment. Coupling alone changes $76.2 \pm 22.3$ to $76.9 \pm 20.9$, which is effectively a wash. The completed `DiagKLA1-zi` experiment combines coupling with isotropic gain initialization and obtains $80.1 \pm 13.3$, below the simpler `DiagKLA-zi` result of $83.7 \pm 10.4$. Thus this parameterization has already been tested: it neither replaces isotropic initialization nor improves on free additive process noise once that initialization is used.

## 5 Recommended Ablation

The transition-coupled process-noise comparison is already complete:

| Variant | Free additive $\omega$ | $(1 - \alpha^2)\rho$ |
|---|---|---|
| Random anisotropic initialization | $76.2 \pm 22.3$ | $76.9 \pm 20.9$ |
| Isotropic gain initialization | $83.7 \pm 10.4$ | $80.1 \pm 13.3$ |

The remaining experiment should isolate covariance-scale choices while keeping free additive $\omega$ and isotropic gain initialization fixed:

| Variant | $\mu$ | $r_t$ | $\omega_t$ |
|---|---|---|---|
| A | learned | learned positive | free additive |
| B | fixed at 1 | learned positive | free additive |
| C | fixed at 1 | fixed at 1 | free additive |
| D | fixed at 1 | bounded relative | free additive |

Run at least the same $5 \times 5$ initialization–data grid. Report mean, total standard deviation, initialization variation, non-finite runs, and convergence time. During training, log $\omega$, $r$, effective write strength, gain anisotropy, the minimum channel gain, and gradient norms for the process-noise, observation-noise, prior, and transition parameters.

If C matches or improves A, the production model should remove both $r_{\text{proj}}$ and $\mu_{\text{param}}$. If D improves C, retain only bounded relative observation reliability. Only after fixing this gauge is a separate uncertainty-parameter learning-rate sweep readily interpretable.

## 6 Why DiagKLA Does Not Yet Dominate IsoKLA

The available implementation tests establish that the Triton kernels reproduce the intended PyTorch recurrence; they do not establish that this recurrence is the best approximation for an associative memory. Empirically, the extra diagonal uncertainty state has not produced a general advantage over the isotropic state. This section separates the approximation question from the kernel-correctness question and states the experiment needed to identify the useful part of the diagonal model.

### 6.1 Current evidence

The controlled 220M/20B information-scale runs do not establish a general DiagKLA advantage. The best default-calibrated variants give:

| Metric | DiagKLA, $s = d_k$ | IsoKLA, $s = 1$ |
|---|---|---|
| WikiText word perplexity ↓ | 28.99 | 28.79 |
| LAMBADA perplexity ↓ | 35.12 | 35.66 |
| Five-task multiple-choice average ↑ | 0.472 | 0.477 |
| S-NIAH-2 at 4K ↑ | 0.272 | 0.046 |
| S-NIAH-3 at 2K ↑ | 0.004 | 0.008 |
| MK-NIAH-1 at 4K ↑ | 0.178 | 0.182 |

General language-modeling quality is mixed and effectively tied. DiagKLA has a large, specific S-NIAH-2 advantage, a direction that also appears in the earlier 220M comparison, but it does not improve the harder single-key or multikey probes. The recent sweeps use one training seed and different GPU world sizes, and a current-code 750M DiagKLA evaluation is not yet available. The evidence therefore supports a task-specific single-key retrieval signal, not a general DiagKLA ranking above IsoKLA.

The noisy-MQAR grid gives a similar result. Uniform process-noise initialization raises DiagKLA from $76.2 \pm 22.3$ to $83.7 \pm 10.4$, while the best stabilized IsoKLA variant obtains $83.0 \pm 8.4$. These are effectively tied at this resolution. The older trained-gain probe that reports large off-key gain and over-writing predates the current initialization and memory kernel; it motivates the mechanism test below but should not be treated as a diagnosis of the current default.

### 6.2 Approximation and calibration

With predicted diagonal covariance $\widehat{\boldsymbol{P}}_t = \mathrm{diag}(\widehat{\boldsymbol{p}}_t)$, the exact Gaussian measurement update produces a dense posterior whose diagonal is

$$p_{t,i}^{\mathrm{exact}} = \widehat{p}_{t,i} - \frac{\widehat{p}_{t,i}^{2} k_{t,i}^{2}}{r_t + \sum_j \widehat{p}_{t,j} k_{t,j}^{2}}. \tag{14}$$

The scan-compatible DiagKLA state instead uses

$$p_{t,i}^{\mathrm{proj}} = \left( \widehat{p}_{t,i}^{-1} + s \frac{k_{t,i}^{2}}{r_t} \right)^{-1}, \qquad s = d_k \quad \text{by default.} \tag{15}$$

Even at $s = 1$, Equation (15) is not the diagonal of the exact posterior in Equation (14). For $s \neq 1$, the gain and information update additionally correspond to different measurement scales: the current gain retains precision $1/r_t$, while the state used by future gains receives precision $s/r_t$. Thus $s = d_k$ is a dimension calibration of a gain-conditioning state, not an exact Kalman consequence.

For a spread-out unit key, $\mathbb{E}[k_i^2] = 1/d_k$, so $s = d_k$ gives a unit expected per-coordinate increment. Averaging these calibrated coordinates recovers the IsoKLA scale,

$$\frac{1}{d_k}\sum_i d_k \frac{k_{t,i}^2}{r_t} = \frac{\|\boldsymbol{k}_t\|_2^2}{r_t} = \frac{1}{r_t}. \tag{16}$$

Matching mean information does not match covariance or gain because inversion is nonlinear. Moreover, for a uniformly distributed unit key,

$$\mathrm{Var}(d_k k_i^2) = \frac{2(d_k - 1)}{d_k + 2} \longrightarrow 2. \tag{17}$$

The coordinate noise therefore does not vanish with width. DiagKLA can interpret coordinate energy fluctuations as persistent uncertainty structure. The projection is also basis dependent: $(1,1)/\sqrt{2}$ and $(1,-1)/\sqrt{2}$ are orthogonal but have identical elementwise squares and hence identical information increments. Whether the learned key basis aligns this diagonal statistic with useful uncertainty is an empirical question.

### 6.3 Information scale separates present writing from future protection

The role of $s$ is clearest after separating the memory update at token $t$ from the uncertainty state passed to later tokens. Define the key-weighted predicted variance and the *effective write strength*

$$z_t = \sum_i \widehat{p}_{t,i} k_{t,i}^2, \qquad \beta_t^{\mathrm{eff}} = \boldsymbol{k}_t^\top \boldsymbol{\kappa}_t. \tag{18}$$

Contracting the diagonal gain with its own key evaluates $\beta_t^{\mathrm{eff}}$ in closed form,

$$\beta_t^{\mathrm{eff}} = \sum_i k_{t,i} \frac{\widehat{p}_{t,i} k_{t,i}}{r_t + \sum_j \widehat{p}_{t,j} k_{t,j}^2} = \frac{\sum_i \widehat{p}_{t,i} k_{t,i}^2}{r_t + z_t} = \frac{z_t}{r_t + z_t}, \qquad 1 - \beta_t^{\mathrm{eff}} = \frac{r_t}{r_t + z_t}. \tag{19}$$

Positive variances and observation noise give $z_t \geq 0$ and $r_t > 0$, hence $\beta_t^{\mathrm{eff}} \in [0, 1)$.

**Residual contraction.** Write $\widetilde{\boldsymbol{S}}_t = \boldsymbol{D}_t \boldsymbol{S}_{t-1}$ for the decayed memory and $\boldsymbol{e}_t = \boldsymbol{v}_t - \widetilde{\boldsymbol{S}}_t^\top \boldsymbol{k}_t$ for the prediction error that the delta rule corrects, so that the memory recurrence is the rank-one update

$$\boldsymbol{S}_t = \widetilde{\boldsymbol{S}}_t + \boldsymbol{\kappa}_t \boldsymbol{e}_t^\top. \tag{20}$$

Transposing Equation (20) gives $\boldsymbol{S}_t^\top = \widetilde{\boldsymbol{S}}_t^\top + \boldsymbol{e}_t \boldsymbol{\kappa}_t^\top$. Reading the updated memory at the same key, the scalar $\boldsymbol{\kappa}_t^\top \boldsymbol{k}_t = \beta_t^{\mathrm{eff}}$ factors out of the rank-one term:

$$\boldsymbol{S}_t^\top \boldsymbol{k}_t = \widetilde{\boldsymbol{S}}_t^\top \boldsymbol{k}_t + \boldsymbol{e}_t \left(\boldsymbol{\kappa}_t^\top \boldsymbol{k}_t\right) = \widetilde{\boldsymbol{S}}_t^\top \boldsymbol{k}_t + \beta_t^{\mathrm{eff}} \boldsymbol{e}_t, \tag{21}$$

$$\boldsymbol{v}_t - \boldsymbol{S}_t^\top \boldsymbol{k}_t = \underbrace{\left(\boldsymbol{v}_t - \widetilde{\boldsymbol{S}}_t^\top \boldsymbol{k}_t\right)}_{=\boldsymbol{e}_t} - \beta_t^{\mathrm{eff}} \boldsymbol{e}_t = \left(1 - \beta_t^{\mathrm{eff}}\right) \boldsymbol{e}_t = \frac{r_t}{r_t + z_t} \boldsymbol{e}_t. \tag{22}$$

Thus $\beta_t^{\mathrm{eff}}$ is the fraction of the current residual corrected at its own key. Equation (22) is the delta-rule instance of the standard Kalman innovation contraction $\boldsymbol{I} - \boldsymbol{K}_t \boldsymbol{H}_t$, which for the scalar-key observation $\boldsymbol{H}_t = \boldsymbol{k}_t^\top$ collapses to the scalar $1 - \boldsymbol{k}_t^\top \boldsymbol{\kappa}_t$. Vanilla DeltaNet is the special case $\boldsymbol{\kappa}_t = \beta_t \boldsymbol{k}_t$ with $\|\boldsymbol{k}_t\|_2 = 1$, which recovers $\beta_t^{\text{eff}} = \beta_t$; here the same quantity is instead an emergent function $z_t/(r_t + z_t)$ of the uncertainty state.

Two properties of Equation (22) are used below. First, the same $\boldsymbol{\kappa}_t$ controls both the value write $+\boldsymbol{\kappa}_t \boldsymbol{v}_t^\top$ and the erasure $-\boldsymbol{\kappa}_t \boldsymbol{k}_t^\top \widetilde{\boldsymbol{S}}_t$, so a gain that assimilates the current observation more strongly necessarily also erases more of what the memory already held at that key. Second, the contraction depends on $\boldsymbol{\kappa}_t$ only through its component along $\boldsymbol{k}_t$: any off-key part $\boldsymbol{\kappa}_{t,\perp}$ satisfies $\boldsymbol{k}_t^\top \boldsymbol{\kappa}_{t,\perp} = 0$ and leaves Equation (22) unchanged while still perturbing the readout at other keys. The same-token residual is therefore blind to exactly the degree of freedom that distinguishes DiagKLA from IsoKLA, which is what Equation (47) is designed to isolate.

To expose causal ordering, temporarily allow a token-specific scale $s_t$. Current DiagKLA first computes $\widehat{\boldsymbol{p}}_t$ and $\boldsymbol{\kappa}_t$ from the incoming state $\boldsymbol{c}_{t-1}$, and only afterward computes the tracked projected covariance below. In this subsection, plain $\boldsymbol{p}_t$ denotes that DiagKLA state, not the exact posterior diagonal.

$$p_{t,i}(s_t) = \left( \widehat{p_{t,i}^{-1}} + s_t \frac{k_{t,i}^2}{r_t} \right)^{-1} = \frac{\widehat{p}_{t,i}}{1 + s_t \widehat{p}_{t,i} k_{t,i}^2 / r_t}. \tag{23}$$

Consequently, for a local intervention that holds the incoming state and token projections fixed,

$$\frac{\partial \boldsymbol{\kappa}_t}{\partial s_t} = 0, \qquad \frac{\partial \boldsymbol{S}_t}{\partial s_t} = 0, \qquad \frac{\partial p_{t,i}}{\partial s_t} = -\frac{k_{t,i}^2}{r_t} p_{t,i}^2 \le 0. \tag{24}$$

The scale at token $t$ therefore does not write that observation harder or change its same-token residual. It only increases the confidence state supplied to the future. In causal order, $\boldsymbol{c}_{t-1}$ determines $\widehat{\boldsymbol{p}}_t$, which determines $\boldsymbol{\kappa}_t$ and $\boldsymbol{S}_t$; $s_t$ then enters $\boldsymbol{c}_t$ and can first affect the memory gain at token $t+1$.

This token-local statement must not be confused with a comparison between two complete runs that use different global values of $s$. Except at the first token, a global $s$ has already changed $\boldsymbol{c}_{t-1}$ and therefore generally changes the current gain. Conditional on fixed projected $\boldsymbol{k}$, $\boldsymbol{\alpha}$, $\boldsymbol{\omega}$, and $r$, its first delayed effect is explicit:

$$\frac{\partial z_{t+1}}{\partial s_t} = -\frac{1}{r_t} \sum_i \alpha_{t+1,i}^2 p_{t,i}^2 k_{t,i}^2 k_{t+1,i}^2 \le 0, \tag{25}$$

$$\frac{\partial \beta_{t+1}^{\text{eff}}}{\partial s_t} = \frac{r_{t+1}}{(r_{t+1} + z_{t+1})^2} \frac{\partial z_{t+1}}{\partial s_t} \le 0. \tag{26}$$

More generally, for $u > t$ the channelwise sensitivity propagates as

$$\frac{\partial p_{u,i}}{\partial s_t} = -\frac{k_{t,i}^2}{r_t} p_{t,i}^2 \prod_{n=t+1}^{u} \frac{\alpha_{n,i}^2}{(1 + s_n \widehat{p}_{n,i} k_{n,i}^2 / r_n)^2}. \tag{27}$$

A larger past scale therefore reduces the later effective correction $\boldsymbol{k}_u^\top \boldsymbol{\kappa}_u$. It need not reduce every component or the norm of an anisotropic $\boldsymbol{\kappa}_u$: reducing the numerator coordinates also reduces their shared denominator. End-to-end trained runs need not be monotone either, because their learned projections can compensate for the changed recurrence.

Equations (26)–(27) also show what is being protected. Suppression is controlled by elementwise-energy overlap such as $\sum_i p_{t,i}^2 k_{t,i}^2 k_{u,i}^2$, not by the signed directional overlap $(\boldsymbol{k}_t^\top \boldsymbol{k}_u)^2$. At fixed projections, a future key with disjoint coordinate support receives no direct effective-gain suppression through this pathway, although the full memory can still differ through intervening writes. Dense orthogonal keys can instead have large energy overlap, including the maximum compatible with orthogonality in the two-dimensional example above. The tracker gives those two keys identical confidence increments and, under a symmetric incoming covariance, identical effective-gain suppression. Their signed memory updates and residuals need not be identical. Large $s$ can consequently suppress a genuinely new orthogonal association rather than selectively preserve only the old one.

Prediction limits the duration of this protection. As $s_t \to \infty$, $p_{t,i} \to 0$ wherever $k_{t,i} \neq 0$, but

$$\widehat{p}_{t+1,i} = \alpha_{t+1,i}^2 p_{t,i} + \omega_{t+1,i} \longrightarrow \omega_{t+1,i}. \tag{28}$$

Positive process noise restores a floor on uncertainty and hence future plasticity; small $\alpha$ also rapidly attenuates the confidence sensitivity in Equation (27) while directly decaying the memory. Thus larger $s$ does not store more content or create capacity by itself. It changes the retention–plasticity operating point until $\boldsymbol{\omega}$ and $\boldsymbol{\alpha}$ wash out that change. If $\boldsymbol{\omega} = 0$ and $\boldsymbol{\alpha} \simeq 1$, an extreme $s$ can instead freeze future gains on the affected coordinates.

The different optima for DiagKLA and IsoKLA are consistent with their different recurrences. For a spread-out normalized key, the DiagKLA increment in one coordinate has expectation $s/(d_k r_t)$, so $s = d_k$ cancels the coordinate dilution without changing the current DiagKLA gain locally. IsoKLA has one scalar state and $\|\boldsymbol{k}_t\|_2^2 = 1$, so its information increment is already $s/r_t$ with no $1/d_k$ dilution. Moreover, the current IsoKLA implementation uses, locally holding the incoming $\widehat{b}_t$ fixed,

$$\beta_t^{\text{iso}}(s) = \frac{\widehat{b}_t}{r_t + s\widehat{b}_t}, \qquad c_t^{\text{iso}}(s) = \widehat{b}_t^{-1} + \frac{s}{r_t}, \qquad \left.\frac{\partial \beta_t^{\text{iso}}}{\partial s}\right|_{\widehat{b}_t} = -\frac{\widehat{b}_t^2}{(r_t + s\widehat{b}_t)^2} < 0. \tag{29}$$

A token-local increase in $s$ in IsoKLA therefore weakens the current write and increases future confidence at the same time. With fixed projections, the lower uncertainty inherited from past larger scales reinforces this reduction in the global-$s$ recurrence. At $s = d_k$ these two anti-plasticity effects can be excessive, whereas DiagKLA obtains delayed protection without the same local write penalty. This gives a mechanistic interpretation of the observed optima $s = d_k$ and $s = 1$, but it is not yet causal evidence that retention alone produces the downstream improvements.

A coherent Gaussian observation-precision multiplier has a different meaning. For a local intervention that holds the predicted prior $\widehat{\boldsymbol{P}}_t$ fixed, increasing precision by $s$ means using $r_t^{\text{eff}} = r_t/s$ in both the current gain and posterior. With $z_t = \boldsymbol{k}_t^\top \widehat{\boldsymbol{P}}_t \boldsymbol{k}_t$, this gives

$$\boldsymbol{\kappa}_t^{\text{coh}}(s) = \frac{s\widehat{\boldsymbol{P}}_t \boldsymbol{k}_t}{r_t + s z_t}, \qquad \beta_t^{\text{coh}}(s) = \frac{s z_t}{r_t + s z_t}, \qquad \frac{\partial \beta_t^{\text{coh}}}{\partial s} = \frac{r_t z_t}{(r_t + s z_t)^2} > 0. \tag{30}$$

The remaining current residual is then $1 - \beta_t^{\text{coh}} = r_t/(r_t + s z_t)$, so a more precise observation is assimilated more aggressively. At the same time, its exact posterior is

$$\boldsymbol{P}_t(s) = \widehat{\boldsymbol{P}}_t - \frac{s\widehat{\boldsymbol{P}}_t \boldsymbol{k}_t \boldsymbol{k}_t^\top \widehat{\boldsymbol{P}}_t}{r_t + s z_t}, \qquad \frac{\partial\, \boldsymbol{\ell}^\top \boldsymbol{P}_t(s) \boldsymbol{\ell}}{\partial s} = -\frac{r_t (\boldsymbol{\ell}^\top \widehat{\boldsymbol{P}}_t \boldsymbol{k}_t)^2}{(r_t + s z_t)^2} \leq 0. \tag{31}$$

Thus coherent higher precision has the two temporally distinct effects expected of a Kalman filter: write the observation harder now, then reduce later corrections in covariance-overlapping directions. Neither the current DiagKLA split nor Equation (29) for $s \neq 1$ has this interpretation. A coherent implementation should use $r_t/s$ once; replacing $r_t$ by $r_t/s$ and also retaining a separate factor $s$ in the information update would double-count the scale and produce an $s^2$ increment.

These mechanisms can be separated explicitly with a current-write scale $s_{\text{write}}$ and a post-write confidence scale $s_{\text{protect}}$:

$$\boldsymbol{\kappa}_t(s_{\text{write}}) = \frac{s_{\text{write}} \widehat{\boldsymbol{p}}_t \odot \boldsymbol{k}_t}{r_t + s_{\text{write}} z_t}, \qquad c_{t,i} = \widehat{p}_{t,i}^{-1} + s_{\text{protect}} \frac{k_{t,i}^2}{r_t}. \tag{32}$$

Current DiagKLA is $(s_{\text{write}}, s_{\text{protect}}) = (1, s)$. Setting both to $s$ makes the scale consistent within the diagonal precision projection, although that projection still is not the exact posterior diagonal. A factorial comparison of $(1,1)$, $(1,d_k)$, $(d_k,1)$, and $(d_k,d_k)$ would separate current assimilation from future protection; because $s_{\text{write}} = d_k$ may saturate the current correction, a gain-matched intermediate value should also be included. Log the same-token $\beta_t^{\text{eff}}$ and residual contraction, the next-token and repeated-key gains, write/erase magnitudes, covariance quantiles, and effective memory half-life. Repeated-key, replacement, and orthogonal-key tasks can then identify whether the benefit comes from retention, impaired rewriting, or diagonal geometry.

A large learned $r_t$ could make the $s = 1$ confidence increment too weak, but the preferred $s = d_k$ does not by itself diagnose observation noise as too large: increasing $r_t$ also weakens the current gain, whereas increasing only $s_{\text{protect}}$ does not. The relevant dimensionless quantities are $z_t/r_t$ for the current write and $s\widehat{p}_{t,i}k_{t,i}^2/r_t$ for the confidence update. These, together with $\omega_{t,i}/\widehat{p}_{t,i}$, should be measured before attributing the result to $r_t$. Likewise, if $r_t$ is freely learned, a globally coherent $s$ is largely reparameterizable into $r_t/s$; the present decoupled scale is not redundant because it changes protection without changing that observation's same-token write locally.

### 6.4 The delta-rule limit and hard writes

Equation (22) specializes cleanly to vanilla DeltaNet, which is the case $\boldsymbol{D}_t = \boldsymbol{I}$ and $\boldsymbol{\kappa}_t = \beta_t \boldsymbol{k}_t$ with $\|\boldsymbol{k}_t\|_2 = 1$, hence $\beta_t^{\text{eff}} = \beta_t$. Reading the updated memory at the key just written is then a convex interpolation between the old readout and the new value,

$$\boldsymbol{S}_t^\top \boldsymbol{k}_t = (1-\beta_t)\boldsymbol{S}_{t-1}^\top \boldsymbol{k}_t + \beta_t \boldsymbol{v}_t. \tag{33}$$

So $\beta_t$ is literally an overwrite fraction: $\beta_t = 0$ ignores the token, $\beta_t = 1$ replaces it, and intermediate values blend. The quantity $\beta_t^{\text{eff}}$ generalizes exactly this scalar.

**The hard write $\beta_t = 1$.** At $\beta_t = 1$ the residual in Equation (22) vanishes identically, so the value is recovered exactly whatever the memory previously held:

$$\boldsymbol{S}_t^\top \boldsymbol{k}_t = \boldsymbol{v}_t, \qquad \boldsymbol{S}_t = \left(\boldsymbol{I} - \boldsymbol{k}_t\boldsymbol{k}_t^\top\right)\boldsymbol{S}_{t-1} + \boldsymbol{k}_t\boldsymbol{v}_t^\top. \tag{34}$$

Because $\|\boldsymbol{k}_t\|_2 = 1$, the matrix $\boldsymbol{I} - \boldsymbol{k}_t\boldsymbol{k}_t^\top$ is idempotent: it is the orthogonal projector onto $\boldsymbol{k}_t^\perp$. A hard write therefore does not attenuate the old content along $\boldsymbol{k}_t$; it annihilates it and writes $\boldsymbol{v}_t$ into the vacated direction.

Equation (34) is the Frobenius projection of $\boldsymbol{S}_{t-1}$ onto the affine set $\{\boldsymbol{S} : \boldsymbol{S}^\top \boldsymbol{k}_t = \boldsymbol{v}_t\}$, that is, a single Kaczmarz step, and general $\beta_t$ is the relaxed variant. This also fixes the stability range: the spectrum of $\boldsymbol{I} - \beta_t\boldsymbol{k}_t\boldsymbol{k}_t^\top$ is $\{1^{(d_k-1)}, 1-\beta_t\}$, so the map is non-expansive exactly for $\beta_t \in [0,2]$, with $\beta_t = 1$ the clean projection and $\beta_t = 2$ a Householder reflection.

**Cross-talk is the price.** Exact recall at $\boldsymbol{k}_t$ is bought by perturbing every non-orthogonal key. Applying Equation (34) to an earlier key $\boldsymbol{k}_u$ gives

$$\boldsymbol{S}_t^\top \boldsymbol{k}_u = \boldsymbol{S}_{t-1}^\top \boldsymbol{k}_u + \langle \boldsymbol{k}_t, \boldsymbol{k}_u \rangle\, \boldsymbol{e}_t, \tag{35}$$

a rank-one corruption in the innovation, scaled by the *signed* inner product and vanishing if and only if $\boldsymbol{k}_t \perp \boldsymbol{k}_u$. Lossless storage of $d_k$ associations therefore requires an orthogonal key set and degrades with the off-diagonal mass of the key Gram matrix.

This sharpens the contrast drawn above. DeltaNet's readout corruption is governed by the signed overlap $\langle \boldsymbol{k}_t, \boldsymbol{k}_u \rangle$, whereas DiagKLA's confidence suppression in Equations (26)–(27) is governed by the unsigned elementwise energy $\sum_i p_{t,i}^2 k_{t,i}^2 k_{u,i}^2$. The pair $(1,1)/\sqrt{2}$ and $(1,-1)/\sqrt{2}$ separates the two: DeltaNet writes one without disturbing the readout of the other, while DiagKLA assigns them identical confidence increments and identical effective-gain suppression.

**What a hard write means for the filter.** Since $\beta_t^{\text{eff}} = z_t/(r_t + z_t)$, the value $\beta_t^{\text{eff}} = 1$ requires $r_t = 0$ or $z_t \to \infty$. A DeltaNet hard write is thus the noiseless-observation, uninformative-prior limit of the filter: the observation is trusted completely and the prior carries no information, while $\beta_t < 1$ corresponds to $r_t > 0$. DeltaNet's learned $\beta_t$ is in this sense a free reparameterization of $z_t/(r_t + z_t)$ that skips the uncertainty state entirely, which is precisely the design choice KLA is contesting.

The consequence is a capability difference, not merely a parameterization difference. Because DeltaNet projects $\beta_t$ directly from the input, token $10^4$ can overwrite as hard as token 10. In KLA the same quantity is derived from an accumulating state, and by Section 7 that state is monotone in the absence of process noise: at $\boldsymbol{\omega} = 0$ the effective write strength can only decrease, so a late hard write is unreachable regardless of what the projections learn. Positive $\boldsymbol{\omega}$ is what buys back DeltaNet's ability to overwrite late in a sequence.

Even with $\boldsymbol{\omega} > 0$ the current model does not operate near a hard write. The stationary point of Equation (55) at the present initialization is $\beta_\infty^{\text{eff}} \simeq 0.62$, a permanent partial-blend regime; reaching $\beta^{\text{eff}} \simeq 1$ requires $z_t \gg r_t$, hence either $r_t \to r_{\min}$ or a large $\boldsymbol{\omega}$. The gauge of Section 2 does not restrict this: choosing $\lambda = 1/r$ maps any learned-$r$ model to an equivalent one with $r \equiv 1$ and $\boldsymbol{\omega} \mapsto \boldsymbol{\omega}/r$, so fixing the covariance unit relocates the scale into $\boldsymbol{\omega}$ rather than capping the reachable overwrite strength. At $r \equiv 1$, for instance, $\beta_\infty^{\text{eff}} \simeq 0.95$ corresponds to $\omega \simeq 18$.

### 6.5 An exact posterior-diagonal alternative

Although the full posterior is dense, its diagonal after one measurement can be computed without materializing a $d_k \times d_k$ covariance. Define

$$D_t = r_t + \sum_j \widehat{p}_{t,j} k_{t,j}^2, \qquad q_{t,i} = \widehat{p}_{t,i} k_{t,i}^2. \tag{36}$$

The denominator $D_t$ is already required for the diagonal Kalman gain. The exact posterior diagonal from Equation (14) can therefore be updated as

$$p_{t,i}^{\text{mom}} = \widehat{p}_{t,i} \frac{D_t - q_{t,i}}{D_t} = \widehat{p}_{t,i} \left(1 - \kappa_{t,i} k_{t,i}\right), \qquad \kappa_{t,i} = \frac{\widehat{p}_{t,i} k_{t,i}}{D_t}. \tag{37}$$

This costs $O(d_k)$ state and arithmetic per token. The multiplicative form in Equation (37) is preferable to subtracting two nearly equal covariance terms. Since $D_t - q_{t,i} = r_t + \sum_{j \neq i} q_{t,j} > 0$, it also makes positivity explicit. Equivalently, the marginal information update is

$$\frac{1}{p_{t,i}^{\text{mom}}} = \frac{1}{\widehat{p}_{t,i}} + \frac{k_{t,i}^2}{r_t + \sum_{j \neq i} \widehat{p}_{t,j} k_{t,j}^2}. \tag{38}$$

Unlike the current update, this expression retains the competition from the other key coordinates instead of replacing its denominator by $r_t$.

The word "exact" applies only to the diagonal of this one posterior when the predicted prior is diagonal. The same update creates off-diagonal entries

$$P_{t,ij} = -\frac{\widehat{p}_{t,i} \widehat{p}_{t,j} k_{t,i} k_{t,j}}{D_t}, \qquad i \neq j. \tag{39}$$

These entries affect the next gain. Discarding them and carrying only $\boldsymbol{p}_t^{\mathrm{mom}}$ is therefore a moment-projected, diagonal assumed-density filter, not the exact multi-token Kalman filter. Exact filtering across tokens requires the dense $\boldsymbol{P}_t$, with $O(d_k^2)$ state and arithmetic per token, as in ExactKLA.

This alternative also changes the parallelization structure. All channels in Equation (37) are coupled through $D_t$, so the recurrence does not decompose into the independent $2 \times 2$ Möbius maps used by the current gain scan. The straightforward implementation is a token-sequential $O(Td_k)$ gain recurrence, preferably fused in Triton; after the gains are produced, the existing chunked Kalman memory kernel remains applicable. Thus the exact posterior-diagonal update saves the dense $O(d_k^2)$ covariance but trades away the current logarithmic-depth prefix scan for the gain.

For the moment projection, the coherent scale from Subsection 6.3 specializes to $r_t^{\mathrm{eff}} = r_t/s$ and gives

$$D_t^{(s)} = \frac{r_t}{s} + \sum_j \widehat{p}_{t,j} k_{t,j}^2, \qquad \kappa_{t,i}^{(s)} = \frac{\widehat{p}_{t,i} k_{t,i}}{D_t^{(s)}}, \qquad p_{t,i}^{(s)} = \widehat{p}_{t,i} - \frac{\widehat{p}_{t,i}^2 k_{t,i}^2}{D_t^{(s)}}. \tag{40}$$

This is the exact-posterior-diagonal specialization of Equations (30)–(31). A direct ablation should compare the current Möbius precision scan against this recurrent moment projection first with $s = 1$, and then, separately, with the coherent calibrated choice $r_t^{\mathrm{eff}} = r_t/d_k$.

**Is the moment projection likely to be a better mixer?** It is a better approximation to the one-step posterior covariance, but it is not necessarily a better drop-in strategy for associative memory. Consider the symmetric case $\widehat{p}_{t,i} = \widehat{p}_t$ and $k_{t,i}^2 = 1/d_k$, and define the effective write strength

$$\beta_t = \frac{\widehat{p}_t}{r_t + \widehat{p}_t}. \tag{41}$$

The current $s = d_k$ precision projection and the moment projection then give

$$p_t^{\mathrm{proj},d_k} = \widehat{p}_t(1 - \beta_t), \qquad p_t^{\mathrm{mom}} = \widehat{p}_t \left(1 - \frac{\beta_t}{d_k}\right). \tag{42}$$

One scalar measurement correctly removes only rank-one uncertainty, so the moment projection reduces each marginal variance by only a $\beta_t/d_k$ fraction. The current rule reduces every coordinate by a $\beta_t$ fraction, or $d_k$ times as much total trace in this symmetric example. It therefore acts as a strong confidence regularizer: future gains are smaller and existing associations are overwritten less aggressively. Indeed, the current $s = 1$ projection also gives smaller posterior variance than the moment projection,

$$p_t^{\mathrm{proj},1} = \frac{\widehat{p}_t d_k r_t}{d_k r_t + \widehat{p}_t} < p_t^{\mathrm{mom}}. \tag{43}$$

Because the empirical information-scale sweep found $s = d_k$ substantially better than $s = 1$, an unretuned moment projection is likely to retain still more uncertainty, produce larger future write/erase gains, and increase interference. It may nevertheless help tasks that require rapid adaptation or frequent replacement of old associations. Posterior faithfulness and memory quality are therefore distinct hypotheses.

The process-noise scale makes a raw comparison especially unfavorable to the moment variant. Prediction injects $O(d_k\omega_t)$ total diagonal uncertainty per token, whereas a rank-one moment update removes only $O(1)$ total uncertainty. Near $\alpha_t^2 = 1$, matching the stationary effective gain of the current $s = d_k$ rule requires approximately

$$\omega_t^{\mathrm{mom}} \simeq \frac{\omega_t^{\mathrm{proj}}}{d_k}, \tag{44}$$

or, equivalently at the level of the stationary covariance-to-noise ratio, an observation-noise scale roughly $d_k$ times larger. Without this recalibration, the current initialization $\omega_t \simeq \omega_{\min} + \text{softplus}(0)$ can drive the moment filter toward high uncertainty and $\beta_t^{\text{eff}}$ close to one.

The appropriate comparison is therefore four-way:

1. the production Möbius projection with $s = d_k$;
2. the same projection with $s = 1$;
3. the recurrent moment projection with the same $\omega/r$ parameterization;
4. the recurrent moment projection with $\omega$ initialized and floored at approximately $1/d_k$ of the production scale, or otherwise calibrated to match the initial and stationary $\beta_t^{\text{eff}}$ distribution.

The raw comparison measures the behavior of each formula under identical parameters; the gain-matched comparison isolates posterior geometry from overwrite temperature. Both should log $\beta_t^{\text{eff}}$, covariance and process-noise quantiles, effective memory half-life, and write/erase magnitude. Repeated-key and orthogonal-key synthetic tasks should precede language model pretraining, because they directly separate rapid rewriting from protection against cross-talk. The moment projection should replace the current rule only if it improves these matched comparisons enough to compensate for losing the parallel Möbius gain scan.

### 6.6 Anisotropic write direction and initialization

For a unit key, decompose

$$\beta_t^{\text{eff}} = \boldsymbol{k}_t^\top \boldsymbol{\kappa}_t, \qquad \boldsymbol{\kappa}_{t,\perp} = \boldsymbol{\kappa}_t - \beta_t^{\text{eff}} \boldsymbol{k}_t, \qquad \boldsymbol{k}_t^\top \boldsymbol{\kappa}_{t,\perp} = 0. \tag{45}$$

IsoKLA enforces $\boldsymbol{\kappa}_{t,\perp} = 0$: uncertainty changes the strength of a write while preserving the binding direction. DiagKLA can rotate both the write and erase directions away from the current key. This can be useful uncertainty preconditioning, or it can write the value into unrelated key directions and create cross-talk. Greater covariance expressivity alone does not distinguish these interpretations.

The current initialization is also not an exact IsoKLA fallback. Zeroing the output map of the process-noise projection makes $\omega_{t,i}$ uniform, but

$$\widehat{p}_{t,i} = \alpha_{t,i}^2/\mu + \omega_t \tag{46}$$

is already channel dependent when $\alpha_{t,i}$ is channel dependent. After one observation, $d_k k_{t,i}^2/r_t$ further leaves the isotropic subspace. The initialization is a useful stabilizer, but the model cannot choose an exact IsoKLA function and add anisotropy only when it is beneficial.

### 6.7 Decisive ablation

The cleanest next experiment holds the diagonal information tracker and effective write strength fixed while intervening only on the gain direction:

$$\boldsymbol{\kappa}_t^{(\eta)} = \beta_t^{\text{eff}} \boldsymbol{k}_t + \eta \boldsymbol{\kappa}_{t,\perp}, \qquad \eta \in \{0, 1\}. \tag{47}$$

Here $\eta = 0$ removes only the off-key component and $\eta = 1$ recovers the current DiagKLA. A gain for $\eta = 0$ would implicate the off-key direction; parity would show little measurable benefit from that direction; a robust gain for $\eta = 1$ would provide direct evidence that anisotropic writes add useful capacity. The comparison should log $\beta^{\text{eff}}$, gain angle, $\boldsymbol{p}$ and $\boldsymbol{c}$ quantiles, $\omega/r$, and cold-channel fraction on current checkpoints.

Two supporting controls are needed:

1. Compare the projected information state in Equation (15) with the sequential exact-posterior diagonal in Equation (14) on a short synthetic task, including orthogonal keys with identical elementwise squares.

2. Run a matched three-seed 220M comparison of DiagKLA, IsoKLA, and the stabilized IsoKLA variant with identical world size, data order, and parameter count. Validate the S-NIAH-2 advantage across context lengths and needle depths before spending on a new large-scale run.

A scale point above the current boundary is also warranted. Test $2d_k$ first and add $4d_k$ only if performance continues to improve. This is an overwrite-temperature diagnostic: larger $s$ increases the post-write confidence state. A token-local change does not strengthen that token's write, while a global change alters later current writes through their covariance prefix. The sweep can locate a better retention–plasticity operating point, but it cannot resolve the posterior inconsistency or establish that off-key anisotropy is useful.

# 7 The Zero-Process-Noise Limit

Because $\boldsymbol{\omega}$ carries part of the covariance-scale symmetry of Section 2 and is what makes the information prediction intractable in closed form, it is natural to ask whether the degenerate choice $\boldsymbol{\omega}_t \equiv 0$ is preferable. It is not. The limit buys a cheaper scan, does not improve the fidelity of the diagonal approximation at all, and removes the plasticity floor that makes the layer a memory rather than an estimator.

## 7.1 It does not restore exactness

Prediction never creates off-diagonal covariance, at any $\boldsymbol{\omega}$. With $\boldsymbol{D}_t = \operatorname{diag}(\boldsymbol{\alpha}_t)$ and $\boldsymbol{\Omega}_t = \operatorname{diag}(\boldsymbol{\omega}_t)$,

$$\widehat{P}_{t,ij} = \alpha_{t,i}\alpha_{t,j}P_{t-1,ij} + \omega_{t,i}\delta_{ij}, \tag{48}$$

so the predict step rescales the existing sparsity pattern and adds only to the diagonal. The measurement is the sole source of off-diagonal mass, and by Equation (39) it contributes $P_{t,ij} = -\widehat{p}_{t,i}\widehat{p}_{t,j}k_{t,i}k_{t,j}/D_t$, which vanishes for every $i \neq j$ if and only if $k_{t,i}k_{t,j} = 0$. The exactness condition for the diagonal family is therefore a condition on the keys alone,

$$\boldsymbol{P}_t \text{ diagonal for all } t \iff \text{every } \boldsymbol{k}_t \text{ is supported on a single coordinate}, \tag{49}$$

and process noise does not appear in it. Setting $\boldsymbol{\omega}_t = 0$ leaves the approximation error of the diagonal state exactly where it was.

What the limit does make exact is the *information* prediction: for invertible $\boldsymbol{D}_t$ it becomes the conjugation $\widehat{\boldsymbol{C}}_t = \boldsymbol{D}_t^{-\top}\boldsymbol{C}_{t-1}\boldsymbol{D}_t^{-1}$ rather than an inverse of a sum. That is a statement about the parameterization of an already-diagonal state, not about how faithfully that state represents the true posterior. The two senses of "exact" should not be conflated.

## 7.2 What the limit does buy

The per-channel information map is linear fractional, with matrix

$$\boldsymbol{M}_t = \begin{pmatrix} 1 + u_{t,i}\omega_{t,i} & u_{t,i}\alpha_{t,i}^2 \\ \omega_{t,i} & \alpha_{t,i}^2 \end{pmatrix}, \qquad u_{t,i} = s\frac{k_{t,i}^2}{r_t}, \tag{50}$$

composed along the sequence by $2 \times 2$ products, which is what makes the gain scan logarithmic depth. At $\omega_{t,i} = 0$ the lower-left entry vanishes, $\boldsymbol{M}_t$ becomes upper triangular, and the Möbius scan degenerates to the affine recurrence

$$c_{t,i} = \frac{c_{t-1,i}}{\alpha_{t,i}^2} + s\frac{k_{t,i}^2}{r_t}, \tag{51}$$

the ordinary decay-accumulate scan of gated linear attention. This is a real implementation saving, and it is the only advantage of the limit.

### 7.3 Covariance windup

A Kalman filter without process noise is recursive least squares, and it inherits the classical covariance windup of that estimator: information only accumulates, so the gain decays to zero and the layer stops writing. By Equation (22) a vanishing $\beta_t^{\text{eff}}$ leaves the residual untouched, $\boldsymbol{v}_t - \boldsymbol{S}_t^\top \boldsymbol{k}_t \to \boldsymbol{e}_t$, so the write becomes a no-op rather than merely a weak one.

The mechanism is a monotonicity. Written as a product, one step of the tracked recursion is

$$p_{t,i} = \underbrace{\alpha_{t,i}^2}_{\leq 1}\ p_{t-1,i} \cdot \underbrace{\frac{1}{1 + s\,\widehat{p}_{t,i}k_{t,i}^2/r_t}}_{\leq 1}, \tag{52}$$

a product of two factors each at most one, with no third factor. Uncertainty can therefore never grow: $\boldsymbol{\omega}$ is the only term in the entire recursion capable of increasing $\boldsymbol{p}$. The same statement holds for the dense filter in the Loewner order, since $\widehat{\boldsymbol{P}}_t = \boldsymbol{D}_t \boldsymbol{P}_{t-1} \boldsymbol{D}_t^\top \preceq \boldsymbol{P}_{t-1}$ and the measurement is a rank-one positive-semidefinite downdate, and it holds for all three trackers considered here: the $s$-scaled projection of Equation (15), its $s = 1$ case, and the moment projection of Equation (37), whose factor $1 - \kappa_{t,i}k_{t,i}$ lies in $[0, 1)$.

In this implementation the transition is moreover strictly contracting. It is produced as $\alpha = \exp(-\exp(A_{\log})\,\text{softplus}(\cdot))$, whose exponent is strictly negative, so $\alpha_{t,i} \in (0, 1)$ always and the harmonic $\alpha = 1$ regime of recursive least squares is never actually reached. Ignoring measurements, which only accelerate the decay, $p_{t,i}/p_{0,i} = \alpha^{2t} \simeq e^{-2t(1-\alpha)}$, so uncertainty survives a context of length $T$ only in channels satisfying

$$1 - \alpha_{t,i} \lesssim \frac{1}{2T}. \tag{53}$$

At $T = 4096$ this demands $\alpha > 1 - 1.2 \times 10^{-4}$: a channel at $\alpha = 0.99$ retains $p_T/p_0 \sim 10^{-36}$, and one at $\alpha = 0.999$ retains $3 \times 10^{-4}$. Because $\boldsymbol{D}_t$ is simultaneously draining $\boldsymbol{S}_t$, the memory empties and cannot be refilled. The idealized $\alpha = 1$ case is much milder but still terminal: with $\mathbb{E}[k_i^2] = 1/d_k$ it gives $c_{t,i} = \mu + ts/(d_k r)$ and hence

$$\beta_t^{\text{eff}} = \frac{d_k}{d_k(1 + \mu r) + st} \xrightarrow[t \text{ large}]{} \frac{d_k}{st}, \tag{54}$$

which at the default $s = d_k$ is $\beta_t^{\text{eff}} \simeq 1/t$, about $2 \times 10^{-4}$ by token 4096. Larger $s$ freezes the memory proportionally faster, with no floor to stop it.

None of this is a numerical defect. The recursion is the exact filter for the generative model $\boldsymbol{S}_t = \boldsymbol{D}_t \boldsymbol{S}_{t-1} + \mathcal{N}(0, \boldsymbol{\Omega}_t)$, and setting $\boldsymbol{\Omega}_t = 0$ asserts that the memory never changes except by a decay already known to the model. Under that assumption every token is evidence about a single unknown initial condition, so a correct posterior must concentrate and a correct filter must stop updating. Recursive least squares converging is the right answer to the question posed; the question is simply the wrong one for an associative memory, in which unpredictable new content is exactly what $\boldsymbol{\Omega}_t$ represents.

### 7.4 The plasticity floor set by process noise

Positive process noise is precisely what prevents this. With $\alpha \simeq 1$, $s = d_k$, and spread-out unit keys, the tracked variance obeys $p_t = \widehat{p}_t/(1+\widehat{p}_t/r)$ and $\widehat{p}_{t+1} = p_t + \omega$, whose fixed point solves $\widehat{p}_\infty^2 - \omega\widehat{p}_\infty - \omega r = 0$:

$$\widehat{p}_\infty = \frac{\omega + \sqrt{\omega^2 + 4\omega r}}{2} = \Theta\big(\sqrt{\omega r}\big) \quad (\omega \ll r), \qquad \beta_\infty^{\text{eff}} = \frac{\widehat{p}_\infty}{r + \widehat{p}_\infty}. \tag{55}$$

The uncertainty floor is the geometric mean of process and observation noise. It therefore degrades gracefully rather than abruptly, but it vanishes continuously as $\omega \to 0$, recovering the windup above. Equation (55) is the stationary counterpart of Equation (28), which states the same protection only in the $s \to \infty$ limit at a single token. At the current initialization—$\omega_{\min} = r_{\min} = 0.05$ with zeroed output maps, giving $\omega \simeq r \simeq 0.74$—it evaluates to $\widehat{p}_\infty \simeq 1.20$ and $\beta_\infty^{\text{eff}} \simeq 0.62$, a healthy retention–plasticity operating point rather than a frozen one.

### 7.5 Two things the limit does not do

It does not remove the scale symmetry. Section 2 survives with one fewer parameter: $\boldsymbol{p}_t' = \lambda \boldsymbol{p}_t$, $r_t' = \lambda r_t$, $\mu' = \mu/\lambda$ still leaves $\boldsymbol{\kappa}_t$ invariant. The gauge choice $r \equiv 1$, $\mu \equiv 1$ is still required and is still the recommended fix.

It is also not the intervention that produced the noisy-MQAR improvement. That result came from zeroing the *output map* of the process-noise projection, which makes $\omega_{t,i}$ uniform across channels at $\omega_{\min} + \text{softplus}(0) \simeq 0.74$—isotropic and large, not small. Setting $\boldsymbol{\omega} = 0$ is the opposite intervention, and conflating the two reads that result backwards. The parameterization $\omega_{t,i} = \omega_{\min} + \text{softplus}(f_\omega(\boldsymbol{x}_t))$ with $\omega_{\min} > 0$ makes the limit unreachable by design.

The zero-process-noise filter is the correct estimator only when the associations are fixed for the whole context and nothing must be overwritten, which is exactly the regime in which recursive least squares is optimal. For a streaming memory it is the degenerate corner that $\omega_{\min}$ exists to exclude.

## 8 Conclusion

The diagonal KDN has a genuine common-scale non-identifiability in $(\boldsymbol{\omega}, r, \mu)$, but the completed noisy-MQAR experiments show that random initial anisotropy is the dominant observed instability. The appropriate design is therefore twofold: start the gain isotropically, and fix a covariance unit—preferably $r = 1$ and $\mu = 1$—so the model learns only uncertainty ratios that affect its predictions.

Process noise should be constrained by that gauge, not removed by it. Section 7 shows that $\boldsymbol{\omega}$ is not merely the third carrier of the scale symmetry: it sets the stationary plasticity floor $\Theta(\sqrt{\omega r})$, and without it the layer degenerates into recursive least squares with a gain that decays as $d_k/(st)$. The fidelity of the diagonal approximation, meanwhile, is governed by key geometry through Equation (49) and is unaffected by $\boldsymbol{\omega}$ either way. Fixing $r$ and $\mu$ while keeping a positive, isotropically initialized $\boldsymbol{\omega}$ is therefore the design that removes the flat direction without removing the mechanism.